

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 3342	Course Title: Logic System Design & Computer Organization
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- Students will be able to demonstrate the fundamental concepts associated with digital logic and circuit design and summarise the basic principles of computer organization.

Course Prerequisites:

Topic	Course name	Semester
Basics of Electronics Engineering	Fundamentals of Electrical and Electronics Engineering	2
Basic functions and features of computer	Introduction to IT Systems Lab	1

Course Outcomes

On completion of the course, the students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate number system conversions, binary arithmetic operations, binary coding & logic gates	11	Applying
CO2	Make use of Boolean algebra and the Karnaugh Map for the implementation of logic functions.	13	Applying
CO3	Summarize the basic operational concepts, memory & I/O of a computer system	10	Understanding
CO4	Outline the working of Processing Unit of a Computer System	9	Understanding
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate number system conversions, binary arithmetic operations binary coding & logic gates		
M1.01	Demonstrate conversions from one number system to different number systems	3	Applying
M1.02	Summarize the types of binary codes & alphanumeric codes	2	Understanding
M1.03	Explain one's-complement and two's-complement of number system	1	Understanding
M1.04	Apply arithmetic operations on binary numbers	3	Applying
M1.05	Outline the properties of basic and universal gates	2	Understanding
Contents: Number systems - decimal, binary, octal, hexadecimal number systems - conversion Types of binary codes - binary coded decimal, excess 3 code, gray code -conversion Alphanumeric codes -ASCII code Complements of number systems -1's complement and 2's complement Binary arithmetic - addition, subtraction, subtraction using complement, multiplication and division Introduction to logic gates - AND, OR, NOT, NAND, NOR, EX-OR and EX-NOR operations Universal property - NAND and NOR gates			
CO2	Make use of Boolean algebra and the Karnaugh Map for the implementation of logic functions.		
M2.01	Realization of universal property of NAND and NOR gates.	2	Applying

M2.02	Explain basic theorems of Boolean algebra	2	Understanding
M2.03	Minimize boolean expressions using product of sum and sum of product	2	Applying
M2.04	Make use of K-Map for the implementation of logic functions	3	Applying
M2.05	Explain Combinational circuits	2	Understanding
M2.06	Explain Sequential circuits	2	Understanding
	Series Test – I	1	
Contents: Realization -AND, OR, NOT, EX-OR and EX-NOR using universal gates Laws,theorems and principles -Laws of Boolean algebra and De-Morgan's theorems Minimization of boolean expressions -Sum Of Products (SOP) expression, Product Of Sum (POS) expression - minterm and maxterm. Karnaugh map - 2, 3, 4 variables - K-map reduction - don't cares in K-map Combinational circuits - half adder,half subtractor, MUX, De-MUX Sequential circuit - Flip flops(S-R,J-K,D,T)-basic concept only, decade counter			
CO3	Summarize the basic operational concepts, memory & I/O of a computer system		
M3.01	Describe various functional units of a Computer system	1	Understanding
M3.02	Explain the basic operational concepts of a computer system.	1	Understanding
M3.03	Explain Bus structures	1	Understanding
M3.04	Outline the memory hierarchy with respect to speed, size and cost.	3	Understanding
M3.05	Outline I/O interfacing with Program controlled I/O & Interrupt Driven I/O	3	Understanding
M3.06	Explain the concept of DMA	1	Understanding
Contents: Computer System - Functional units - Basic operational concepts, Bus Structures Memory Hierarchy - Read Only memory- PROM, EPROM, EEPROM, Flash Memory., Primary Memory-static ,dynamic,comparison, Secondary memory -Features, examples Input/Output Organisation - Accessing of I/O devices -Program controlled I/O,Interrupt Driven I/O Direct Memory Access (basic concept only)			
CO4	Outline the working of Processing Unit of a Computer System		
M4.01	Explain the internal architecture of a processor	2	Understanding

M4.02	Explain the Execution of an instruction.	2	Understanding
M4.03	Summarise the various methods to generate control signals	3	Understanding
M4.04	Outline the principle of Instruction pipelining	1	Understanding
M4.05	Describe flynn's classification of processors	1	Understanding
	Series Test – II	1	
Contents: Processing Unit: Von Neumann Architecture, Processor Organization, Register Organization Instruction Cycle: Instruction Fetch -Execute - Interrupts Control unit: Hardwired Control - Microprogrammed Control Instruction pipelining: need of pipelining, 2-stage pipelining Parallel processing: Flynn's classification			

Text / References:

T/R	Book Title / Author
T1	M. Morris Mano and Michael Cilettio - Digital Design - Pearson- 5th Edition
T2	Floyd and Jain -Digital Fundamentals - Pearson- 8th Edition
T3	Carl Hamachar, Computer Organization, McGraw Hill, 5th ed.
T4	William Stallings, Computer Organization and Architecture, Pearson Education , 8th ed
R1	A. Anand Kumar- Fundamentals of digital circuits PHI
R2	John Hayes, Computer Architecture and Organization ,McGraw Hill-1998

Online Resources:

Sl. No.	Website Link
1	https://nptel.ac.in/courses/108106177
2	https://nptel.ac.in/courses/117105080
3	https://nptel.ac.in/courses/106105163
4	https://nptel.ac.in/courses/106108100